

Generative design for a self-balancing unicycle: a re-implementation of the chassis comparison and of the topology-optimisation step

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Abstract

Chen, Cheng and Han (SPIE ACAIB 2022) grew the seven printed brackets of a reaction-wheel unicycle with Fusion 360's generative design and compared the resulting chassis against a conventional one in Simulink under the same PID controllers, reporting a pitch settling time of 0.69 s against 19.20 s. Neither the Simulink model nor the Fusion 360 files are available, so this document re-implements both halves of the paper: (i) the dynamics of the paper's Sec. 3 (a wheeled inverted pendulum for pitch and travel, a reaction-wheel pendulum for roll, brushed DC motors, three PID loops), with the wheel/body coupling terms kept, and (ii) the generative-design step as SIMP topology optimisation with a stress post-check on the paper's three main brackets. With the component masses of the paper's Table 1 and stated estimates for what the paper does not give, the generative chassis (690.00 g, centre of mass 15.00 cm above the axle) settles in 0.70 s in pitch, matching the paper's 0.69 s; the conventional chassis (861.00 g, 12.00 cm) settles in 0.74 s with the same gains. The paper's large gap appears when the fixed gain set is scaled towards the conventional chassis' stability edge: at $\times 1.5$ the generative chassis settles in 0.67 s and the conventional one enters a limit cycle at the voltage limit for the whole 12.00 s run. The physical effect is a slower unstable pole (7.81 s^{-1} against 8.97 s^{-1}) from the higher centre of mass, which buys 10.00 % of gain margin, 5.00 ms of loop-delay tolerance and a slower admissible control period at the same motors. The topology optimisation removes 70.00 % of the side support's mass at a safety factor of 44.80.

Keywords: generative design, topology optimisation, SIMP, self-balancing unicycle, reaction wheel, wheeled inverted pendulum, PID, gain margin.

1 Overview

The paper has two parts. The structural part splits the unicycle body into seven parts by loading condition, runs each through Fusion 360's generative design (preserved regions, obstacles, a starting shape, a material list), selects outcomes on a mass/safety-factor chart, mirrors and merges them, and verifies the merged body in Inventor (safety factor > 15). The control part models the unicycle as two pendulums driven by DC motors, closes three PID loops, and compares the generative and the conventional chassis released from 5.00° on both axes: settling time in a 5.00 % band, the peak movement of the top point, and motor power.

This re-run keeps the paper's protocol and replaces the unavailable tools with open ones: `experiments/unicycle/model.py` for the dynamics, controllers and metrics; `experiments/unicycle/topopt.py` for the generative step; `run.study.py`, `make_figures.py` and `make_videos.py` for the experiments, the figures and the videos of the website page `unicycle.html`. The browser twin (`assets/js/unicycle.sim.js`) is the same model line for line.

2 Method

2.1 Dynamics

Body frame: origin at the drive-wheel axle, x forward, z up. Pitch φ (positive = leaning forward) and travel x form a wheeled inverted pendulum with drive-wheel mass M_w , inertia J_w , radius r , body mass m (everything but the drive wheel), body inertia J_b about its own centre of mass at height l above the axle, and the drive torque τ_1 acting between wheel and body:

$$(J_b + ml^2) \ddot{\varphi} + ml \cos \varphi \ddot{x} - mgl \sin \varphi = -\tau_1, \quad (1)$$

$$(M_w + m + J_w/r^2) \ddot{x} + ml \cos \varphi \ddot{\varphi} - ml \dot{\varphi}^2 \sin \varphi = \tau_1/r. \quad (2)$$

Roll ρ is a reaction-wheel pendulum about the ground contact line, with J_p the inertia of the whole robot about that line (reaction-wheel spin excluded), M_t the total mass, l_b the centre of mass above the ground, J_2 the reaction wheel's spin inertia and θ_2 its angle relative to the body:

$$J_p \ddot{\rho} - M_t g l_b \sin \rho = -\tau_2, \quad J_2 (\ddot{\rho} + \ddot{\theta}_2) = \tau_2. \quad (3)$$

Both motors are brushed DC machines with back-EMF and viscous friction, $\tau = K_T(u - K_E \omega)/R - b\omega$, ω the relative speed the motor sees ($\dot{x}/r - \dot{\varphi}$ for the drive, $\dot{\theta}_2$ for the reaction wheel), $|u| \leq 11.10$ V. Electrical power is $P = ui$ with $i = (u - K_E \omega)/R$; it is negative while regenerating. The paper's Eqs. (3.6)–(3.9) are (1)–(3) with the $ml \cos \varphi$ coupling dropped and the motor term written with K_E in V/rpm (the $60/2\pi$ factor); their 1/2 on the input is not reproduced here.

Assumptions and validity. The two subsystems are decoupled (no gyroscopic coupling between the spinning reaction wheel and the pitch motion, no yaw); the drive wheel is thin, so roll pivots about a line; no ground friction unless stated (a viscous rolling resistance is a parameter, swept in §3); rigid bodies; the body inertia is isotropic. These are the paper's assumptions too. A 60.00° excursion on either axis counts as a fall.

2.2 Controllers

The paper's Fig. 10: three PID loops, one gain set for both chassis. Translation and pitch share the drive motor,

$$u_1 = K_p \varphi + K_d \dot{\varphi} + K_i \int \varphi + K_x x + K_{dx} \dot{x}, \quad u_2 = K_{p,\rho} \rho + K_{d,\rho} \dot{\rho} + K_{i,\rho} \int \rho - K_w \dot{\theta}_2. \quad (4)$$

The re-run's fixed set is $K_p = 180$, $K_d = 12$, $K_i = 40$, $K_x = 18$, $K_{dx} = 22$ (V per rad, rad/s, rad/s, m, m/s), $K_{p,\rho} = 280$, $K_{d,\rho} = 40$, $K_{i,\rho} = 0$, $K_w = -0.18$: a pure angle PID on the reaction wheel cannot hold the wheel speed bounded, so the standard small wheel-speed term is added and the roll integrator dropped; the pitch loop keeps its integrator with an anti-windup clamp at 0.50 rad/s. The controller runs at 200.00 Hz with zero-order hold; the physics is integrated with RK4 at 1.00 kHz; a pure loop delay and the control period are experiment parameters. A second gain set per chassis is obtained by LQR on the linearised plant (weights $\text{diag}(20, 1, 200, 1)$, $R = 0.05$ for pitch; $\text{diag}(200, 1, 10^{-3})$, $R = 0.05$ for roll) and mapped onto the same loop structure.

2.3 Linearisation

About the upright equilibrium the pitch subsystem has state $(x, \dot{x}, \varphi, \dot{\varphi})$ and the roll subsystem $(\rho, \dot{\rho}, \theta_2)$; `model.linearise` returns (A, B) for both with the motor damping $d = K_T K_E / R + b$ included, and is checked against finite differences of the nonlinear model (test). With the motors free, the unstable pitch pole is $\sqrt{mgl(M_w + m + J_w/r^2)/\det \bar{M}}$, essentially $\sqrt{g/l_{\text{eff}}}$: a higher centre of mass falls more slowly.

2.4 Parameters

Component masses are the paper’s Table 1. Everything else is an estimate from the photographs and the CAD render, stated once in `params.py` (Table 1); the conventional chassis follows the paper’s two statements about it (heavier, lower centre of mass) with the structure mass doubled and the centre of mass 3.00 cm lower. The grid sweep of §3.4 shows how the conclusions move with these two.

quantity	generative	conventional	note
printed structure	169.00 g	340.00 g (assumed)	paper: 169.03 g
components	481.00 g	481.00 g	RW 165.6 + 74.3, drive motor 96, batteries 135, boards 10 (paper)
total with the drive wheel	690.00 g	861.00 g	drive wheel 40.00 g, 70.00 mm
body CoM above the axle l	15.00 cm	12.00 cm	above the ground l_b : 17.60 cm, 14.90 cm
body radius of gyration	$0.60 l$	$0.62 l$	
unstable pitch pole	7.81 s^{-1}	8.97 s^{-1}	roll: $6.57, 7.19 \text{ s}^{-1}$
$mg l$ (pitch)	0.96 N m	0.97 N m	nearly equal
reaction wheel	120.00 mm ring, $J_2 = 0.85 \text{ mr}^2$		
drive motor	$K_T = K_E = 0.12$, $R = 2.80 \Omega$, stall 0.48 N m		geared, 12 V class
reaction-wheel motor	$K_T = K_E = 0.045$, $R = 1.50 \Omega$, stall 0.33 N m		direct drive
supply	11.10 V (3S)		

Table 1: Parameters of the re-run. Masses in the first two rows are the paper’s; the rest are estimates.

2.5 Topology optimisation as the generative step

For a bracket under a static load, the core of a generative-design study is compliance-minimising topology optimisation with preserved and forbidden regions. `topopt.py` is the 88-line SIMP formulation of Andreassen et al.: bilinear Q4 plane-stress elements on a 1.00 mm grid, element stiffness scaled by $E_{\min} + x^3(E_0 - E_{\min})$, a density filter of radius $r_{\min}$, an optimality-criteria update with move limit 0.2 and a bisection on the volume constraint, preserved elements pinned at 1 and obstacles at 0, 70 iterations or a change below 0.01. A stress check follows on the final layout: the displacement of the last solve is scaled to Nylon PA12 ($E = 1.70 \text{ GPa}$, yield 48.00 MPa, $\rho = 1010.00 \text{ kg/m}^3$) and a 6.00 mm plate, the von Mises stress at each element centre is scaled by the density, and the safety factor is the yield over the largest stress on elements with density above 0.5. Three load cases mirror the paper’s Fig. 1: the side support (10.00 N from the top plate, 5.00 N sideways both ways, fixed at the base), the reaction-wheel front support (the wheel’s weight, the motor torque as a 6.00 N couple, 2.00 N sideways) and the bottom plate (both ends fixed, a 10.00 N motor, two 5.00 N battery loads, a 3.50 N bearing pull). This is a stiffness-driven layout with a stress post-check, which is also what the paper did (generate for stiffness, verify the safety factor in Inventor).

3 Experiments

All runs are deterministic (`run_study.py control`: 314 simulations; `run_study.py topopt`: 29 optimisations). Metrics: settling time as the first time after which the angle stays within 5.00 % of its

initial value; peak absolute displacement of the top point $x + H \sin \varphi$ ($H = 0.26$ m); mean and peak electrical power; energy as the integral of the positive power.

3.1 The paper's protocol

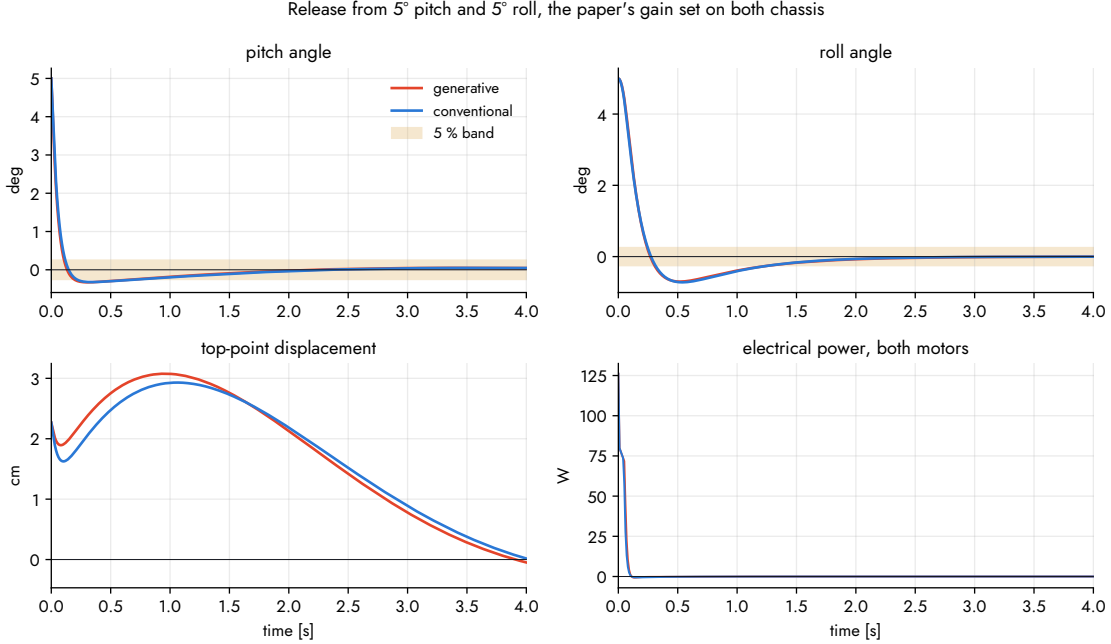


Figure 1: Release from 5.00° pitch and 5.00° roll with the fixed gain set. Pitch settles in 0.70 s (generative) and 0.74 s (conventional); roll in 1.27 s on both, limited by the reaction-wheel motor at the voltage limit for the first 0.20 s; peak power 126.00 W on both (the voltage limit on both motors); energy 5.36 J against 4.90 J.

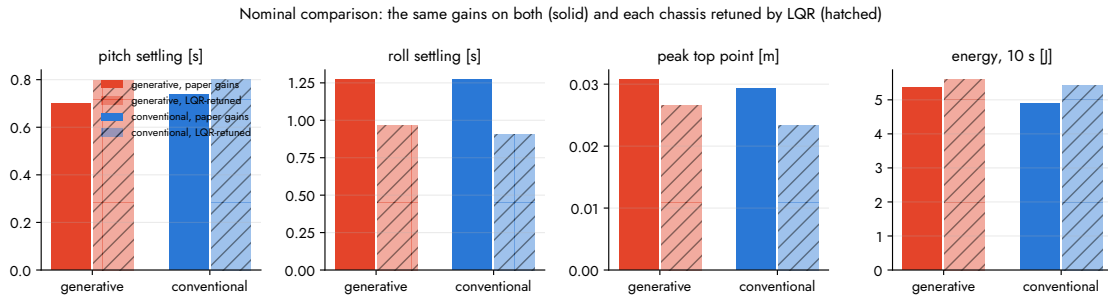


Figure 2: The nominal comparison (solid) and each chassis retuned by LQR (hatched): 0.80 s and 0.80 s in pitch. Retuned, the chassis difference nearly vanishes.

With the paper's protocol the two chassis behave almost alike (Fig. 1): the generative one is 40.00 ms faster in pitch, both saturate the motors at the release, and the conventional one uses slightly less energy because its lower centre of mass stores less potential energy per degree. The generative chassis' 0.70 s matches the paper's 0.69 s; the conventional chassis' 19.20 s of the paper is not reproduced by these gains.

3.2 At the edge: the gain-margin explanation

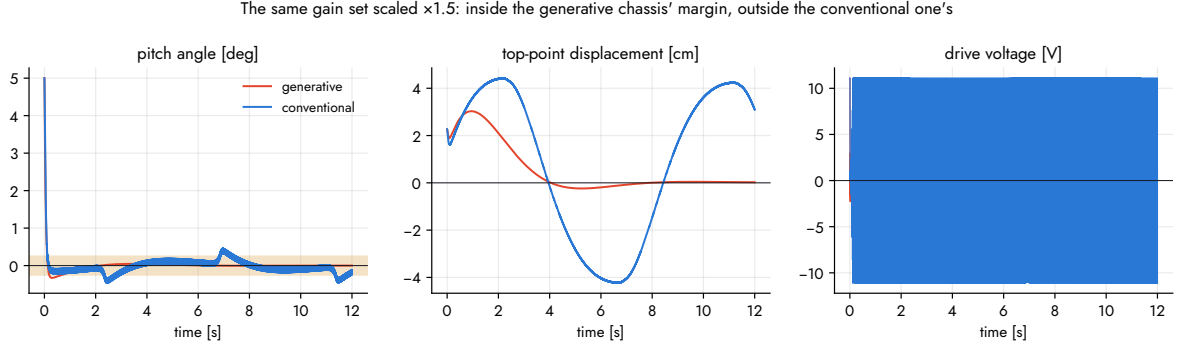


Figure 3: The same gain set scaled $\times 1.5$. The generative chassis settles in 0.67 s; the conventional one enters a limit cycle at the voltage limit: over the last 5.00 s the drive voltage is saturated 81.20 % of the time, the pitch buzzes with an RMS of 0.17° (peak 0.45° , so it dips in and out of the 5.00 % band without settling), the wheel wanders over 8.55 cm, mean power is 43.90 W, and the run uses 528.00 J in 12.00 s against 5.44 J.

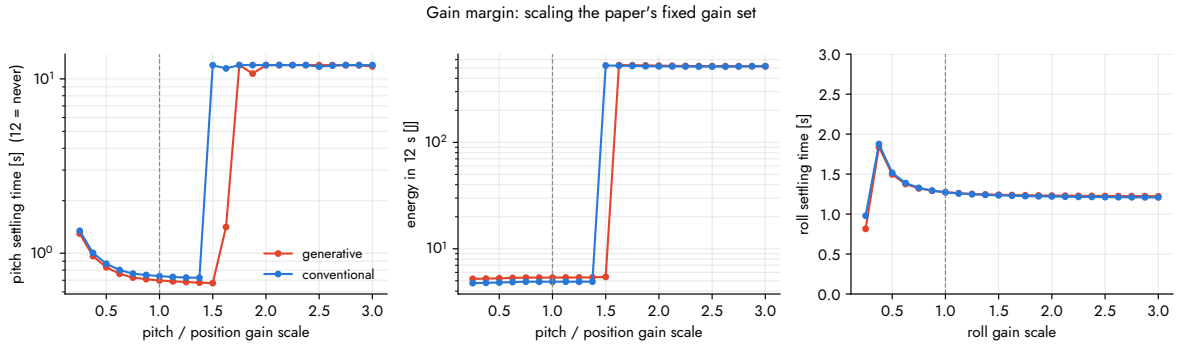


Figure 4: Scaling the pitch/position gains: settling time (left, 12 = never), energy (middle) and the roll loop's own scale (right). The conventional chassis leaves its margin at $\times 1.38$, the generative one at $\times 1.62$.

Scaling the pitch and position gains together (Fig. 4) both chassis improve up to $\times 1.375$; there the conventional chassis breaks into a limit cycle while the generative one holds to $\times 1.5$ (Fig. 3). A gain set inside one chassis' margin and outside the other's produces exactly the paper's pattern: a sub-second settling against a run that never settles. The re-run therefore reads the paper's Table 2 as a gain-margin effect of the fixed gain set, not as a difference in what the two chassis can do when tuned.

3.3 Delay, control period, tilt and push

The slower unstable pole shows up wherever the controller is late (Fig. 5): the generative chassis tolerates 5.00 ms more loop delay and a 10.00 ms longer control period before falling, and starts wandering 10.00 ms later. What the motors set, the chassis cannot move (Fig. 6): the recoverable tilt and push are identical, peak power is the voltage limit on both, and a 5.00° recovery costs

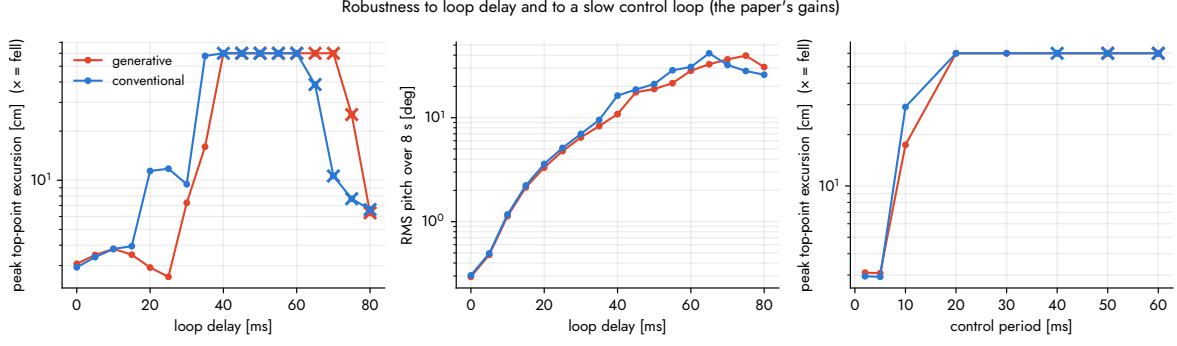


Figure 5: Peak top-point excursion and RMS pitch against a pure loop delay, and against the control period, with the fixed gains. The conventional chassis falls at 40.00 ms of delay and a 40.00 ms period; the generative one at 45.00 ms and 50.00 ms.

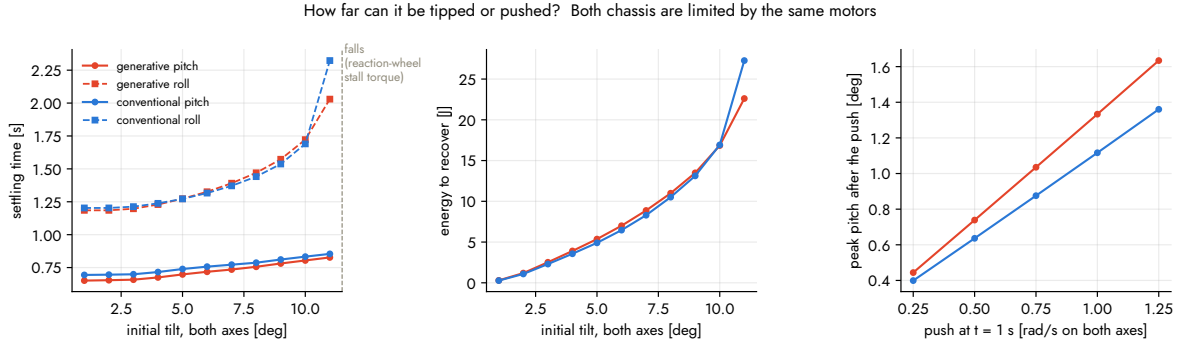


Figure 6: Initial tilt and impulsive pushes. Both chassis recover from up to 11.00° and from pushes up to 1.25 rad/s on both axes; the limit is the reaction-wheel motor's stall torque against $M_t g l_b \sin \rho$, the same on both.

5.00 J either way. Viscous rolling resistance up to 0.40 Ns/m shortens the pitch settling by 10.00 % on both and changes nothing else (friction sweep in out/study.json).

3.4 The (mass, centre of mass) map

Fig. 7 places the two chassis on the map. With these gains any chassis with its centre of mass below about 11.00 cm falls; the conventional chassis sits one row above that edge, the generative one three. The pole (Fig. 8) is set by the height, not the mass; the mass only moves mgl , which is nearly equal for the two.

3.5 Topology optimisation

At the chosen fills the side support keeps a safety factor of 44.80 (solid plate: 46.10) at 13.10 g instead of 43.60 g (−70.00 %); the reaction-wheel support 116.00 at −72.00 %; the bottom plate 35.50 at −68.00 % (Fig. 10, Fig. 11). Stresses are a fraction of a megapascal at these loads: the design is stiffness- and mass-driven, consistent with the paper's safety factor above 15 on the merged body.

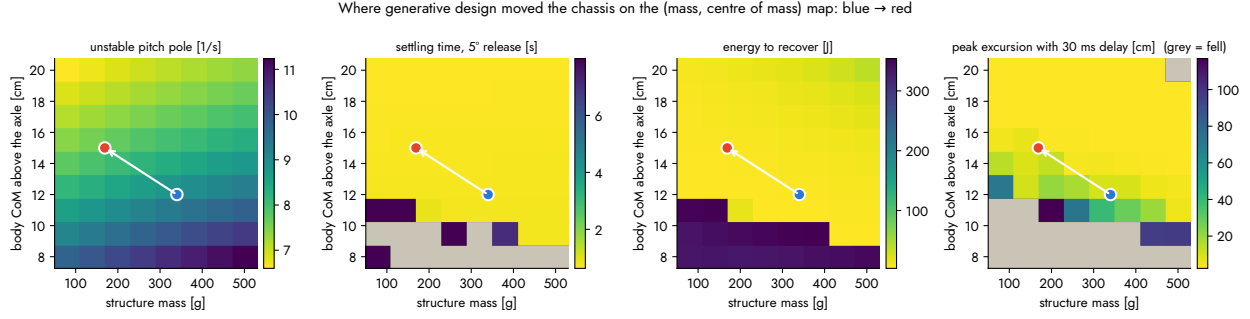


Figure 7: 72 chassis on a grid of structure mass and centre-of-mass height with the fixed gains: unstable pole, settling time, energy, and the peak excursion with 30.00 ms of loop delay (grey = fell). Blue is the conventional chassis, red the generative one.

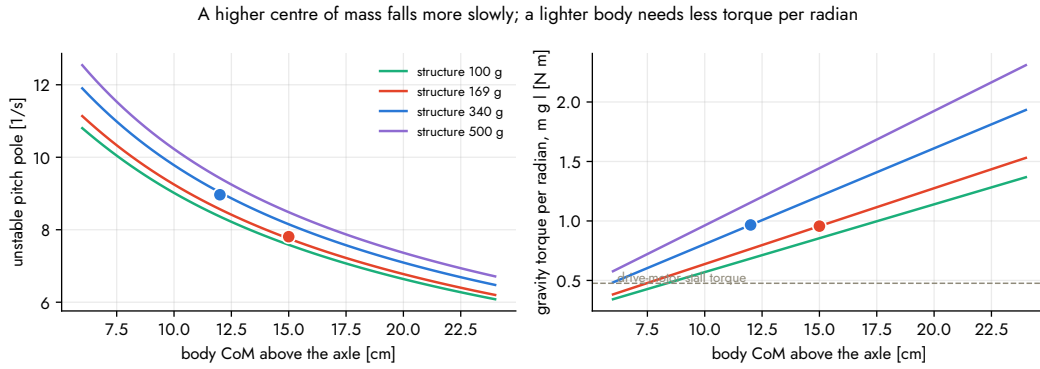


Figure 8: The plant the controller sees: the unstable pitch pole falls with the centre-of-mass height and hardly depends on the structure mass; mgl is what the drive motor's stall torque must beat.

4 Implementation and Usage

experiments/unicycle/ is the package: `params.py` (Chassis, Motor, Gains, every estimate), `model.py` (Unicycle, simulate with pushes, delay and control period, metrics, linearise, lqr), `topopt.py` (LoadCase, topopt, stress_check, CASES). `run_study.py` control|topopt runs the experiments on a process pool and writes out `study.json`, `out/topopt.json` and `out/topopt.npz`; `make_figures.py` writes `docs/figures/*.pdf`, the website PNGs, `assets/data/unicycle_results.json` and `docs/numbers.tex`; `make_videos.py` renders the videos with `render.py`, a small software rasteriser of the 3-D geometry; `tests/` holds six tests (energy conservation of the unactuated pendulum, linearisation against finite differences, open-loop instability and closed-loop stability of both chassis, LQR, the settling-time definition, a small topology case). The website page `unicycle.html` runs the same model in JavaScript (`unicycle_sim.js`, `unicycle_scene3d.js`) and the same SIMP optimiser with a matrix-free conjugate-gradient solve (`unicycle_topopt.js`).

The hardware path: the controller structure is the paper's Arduino-style three-loop PID; the model's motor and inertia parameters are the ones to measure on the printed robot (motor constants from a stall and a no-load test, the centre of mass from a balance, the roll inertia from a pendulum swing), after which the gain-margin and delay sweeps give the loop rate the firmware needs.

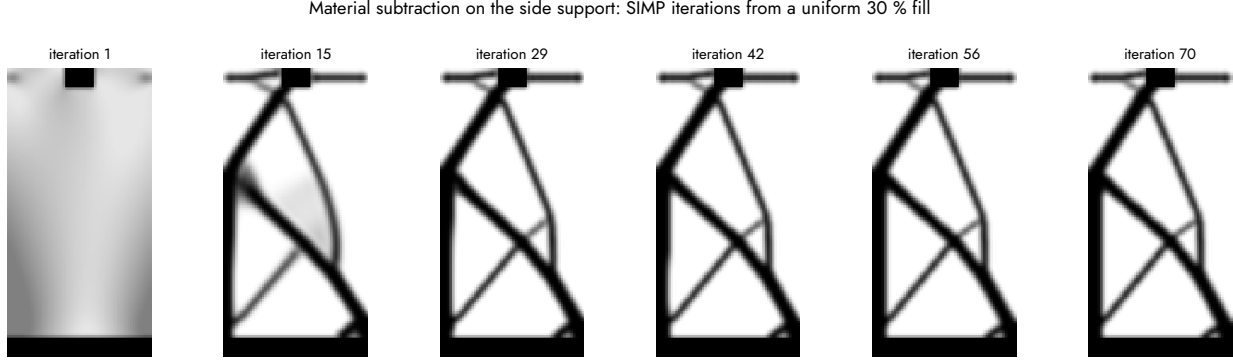


Figure 9: Material subtraction on the side support, from a uniform 30.00 % fill (the paper’s Fig. 3, re-made).

5 Limitations

- The paper’s Simulink/Simscape multibody model and its Fusion 360 outcomes are not reproduced; this is a re-implementation of the same equations with the coupling terms kept, on estimated inertias, motor constants and a conventional chassis defined by two statements. The nominal generative settling time matches the paper; the conventional chassis’ behaviour depends on where the fixed gain set sits relative to its margin.
- The roll settling (1.30 s) is much slower than the paper’s 0.06 s, which would need an implausible reaction-wheel torque; the paper’s power numbers are unscaled Simulink signals and are not compared.
- The pitch and roll subsystems are decoupled; gyroscopic coupling of the spinning reaction wheel, yaw, ground friction and compliance are absent. Sensor noise and quantisation are absent; delay and control period are the only loop imperfections swept.
- The topology optimisation is 2-D plane stress on three brackets with hand-set loads; the paper’s studies were 3-D with seven parts and a materials list. Safety factors are stress post-checks on stiffness-optimised layouts, not stress-constrained designs.
- The pure angle PID of the paper is modified in roll (wheel-speed term, no integrator); without it the reaction wheel winds up to its speed limit.

References

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The three brackets of the chassis after topology optimisation: layout with loads (red) and supports (blue), and stress

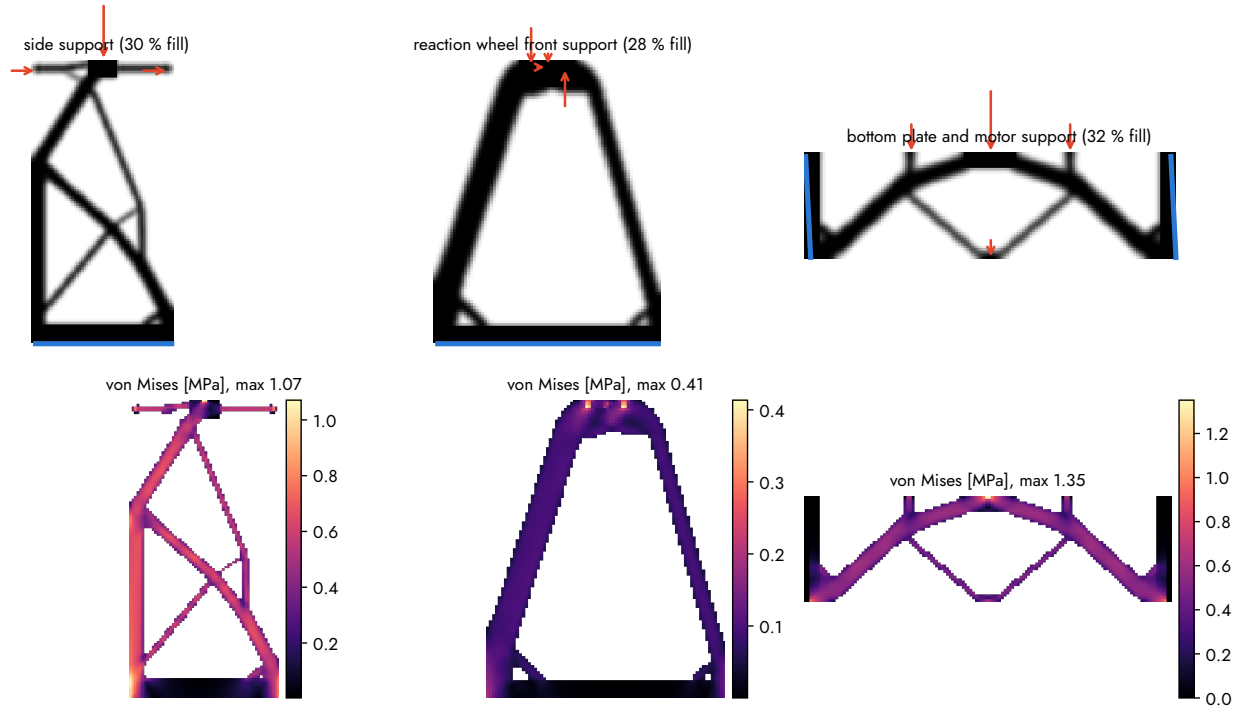


Figure 10: The three brackets at their chosen fills, with loads and supports, and the von Mises stress on the final layouts.

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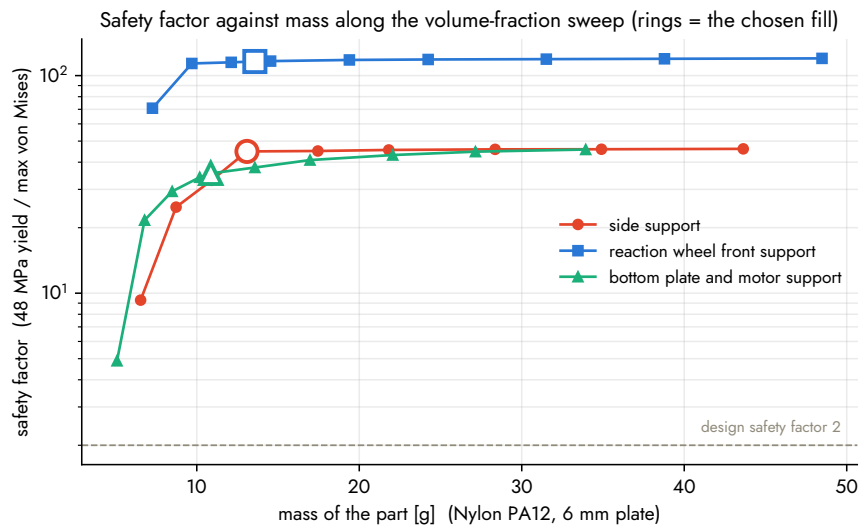


Figure 11: Safety factor against mass along the volume-fraction sweep (the paper's Fig. 5, re-made). Above about 30.00 % fill the safety factor saturates.